DataCopilot

A Reliable Natural-Language Data Analysis System for CSV Files and SQL Databases

Capstone Project – Module 3 Submission: Data Collection and Preparation

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Date	13 November 2026
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DataCopilot – Module 3: Data Collection and Preparation

Nov 13, 2026 · @Mritunjay Sharma

Introduction

Module 3 prepares the data DataCopilot is evaluated on. Five sources were acquired: the BIRD text-to-SQL benchmark and four public datasets chosen to test different difficulties. Each dataset was profiled for quality issues, cleaned with a logged and reproducible pipeline, and explored to understand its structure. On top of the cleaned data, 550 evaluation questions were built: answerable questions with verified gold answers, natural Hinglish versions, and deliberately unanswerable questions in English and Hinglish. Finally, answerability features were engineered for the classifier that will be trained in Module 4.

All scripts, reports, question sets and the EDA notebook are in the GitHub repository. Raw and processed data files are not committed, in line with their licences.

1. Data Sources

The datasets were chosen so that each tests a different difficulty, from clean benchmark data to messy government data.

Dataset	Source	Licence	Size	What it tests
BIRD dev set	bird-bench.github.io (official dev.zip, version dev_20240627)	Research use	1,534 questions, 11 SQLite databases	Main text-to-SQL benchmark, comparable with published results
Olist Brazilian E-commerce	Kaggle (olistbr)	CC BY-NC-SA 4.0	9 linked CSVs, about 99k orders (2016–2018)	Joins across many tables
IPL	Cricsheet	ODC-By 1.0	1,243 matches (2008–2026), ball-by-ball	Large data, Indian names, Hinglish questions
Formula 1	Jolpica-F1 (community successor to Ergast)	CC BY-NC-SA 4.0	18 CSVs, 1950–2026 (filtered to 2014+)	Dates, durations, missing values

Dataset	Source	Licence	Size	What it tests
Crop production	data.gov.in, via a Kaggle mirror	GODL-India (source data)	246,091 rows, 1997–2015	Messy real-world government data

Notes on provenance. The official data.gov.in API repeatedly timed out, so the crop data was taken from a widely used Kaggle mirror of the same government dataset. The mirror is treated as an unofficial copy and credited alongside data.gov.in and the Ministry of Agriculture and Farmers Welfare. The F1 data comes from Jolpica-F1 because the Ergast API shut down in early 2025. The original BIRD dev set was chosen over the cleaned November 2025 split so results stay comparable with published work; the cleaned split is documented as a possible second check.

Full URLs, file lists and row counts are in `docs/planning/data_sources.md`.

2. Data Quality Assessment

A profiling script (`scripts/data_quality.py`) checked every dataset for row and column counts, types, missing values, duplicates, odd column names and suspicious values. One report per dataset is in `docs/data_quality/`. The most important issues are below.

Olist

- 261,831 exact duplicate rows in the geolocation table.
- 2,082 extra city-name spellings that differ only by accents or case (`são paulo` vs `sao paulo`), plus 9 with stray symbols such as `sa£o paulo`.
- 42 geolocation points outside Brazil.
- Typos in column names (`product_name_lenght`).
- Business-rule oddities, reported but not changed: 1,603 shared review IDs, 1,359 orders handed to the carrier before approval, 775 orders with no items, and 9 non-positive payments.

IPL

- 49 deliveries numbered `x.10`. Read as a number, `4.10` becomes `4.1` and collides with a different ball, silently corrupting the data.
- Season labels such as `2007/08` instead of calendar years (42,546 rows).
- Inconsistent team names (`Rising Pune Supergiants` vs `Supergiant`) and 60 venue name variants for 41 grounds.

Formula 1

- 10 boolean columns stored as `t` / `f` text and 4 duration columns stored as `HH:MM:SS` text.
- Status codes stored as integers whose meanings are defined only in the source code.
- Standings stored after every session rather than once per season.

Crop production

- Every `Season` value padded with trailing spaces (`'Kharif '`), as are 7,283 state names. A query filtering on `= 'Kharif'` would silently return nothing.
- 3,730 missing production values (1.5%) and 3,523 rows of exactly zero, which may mean crop failure or missing data.
- No units stated, and coconut counted in nuts rather than tonnes, so totals across crops are not meaningful.
- 2015 is incomplete, with only 562 rows.

Why this matters for DataCopilot. Several issues (padded values, number-like identifiers, inconsistent names) produce SQL that runs without error but returns wrong or empty results. These are exactly the silent failures DataCopilot is designed to expose, and they justify sending sample values to the LLM along with the schema.

3. Data Cleaning and Preparation

A single script (`scripts/prepare_data.py`) cleans all four datasets with shared helper functions. Every step and the number of rows it affected is written to `docs/data_quality/cleaning_log.md` , so the process is reproducible and auditable.

Dataset	Main cleaning steps	Output
Olist	Column names to snake_case; typo fixes; city names normalised (accents, case, symbols); 279,669 geolocation duplicates removed; 33 out-of-country points dropped	SQLite database with 9 indexed tables (`olist.sqlite`)
IPL	Ball numbers split into `over` and `ball_in_over` ; season labels converted to calendar years; team names unified, with franchise columns for renamed teams; 60 venue names reduced to 41	`ipl_deliveries.csv` , `ipl_matches.csv`

Dataset	Main cleaning steps	Output
Formula 1	Text booleans and durations converted to proper types (durations to milliseconds); `is_latest` flag for season-end standings; filtered to 2014+ seasons (276 races)	Six CSVs: races, results, laps, pit stops, driver and team standings
Crop production	Padded text trimmed in all columns; missing production kept as NULL, not imputed; zero values kept and flagged	`crop_production.csv`

Design decisions

- **No imputation.** Missing values stay missing, because invented values would create gold answers that are not in the real data.
- **Flag rather than delete** suspicious records (business-rule oddities, zero production) unless they are clear errors, so the data still reflects reality.
- **Ambiguous merges are left for review**, such as possibly renamed IPL grounds, rather than guessed.
- **Validation.** Cleaned F1 champions for 2014–2024 were checked against the real results, and all matched.

The cleaning code is covered by unit tests on small synthetic data; the full test suite had 170 passing tests at the end of this step.

4. Exploratory Data Analysis

The EDA notebook (`notebooks/03_eda.ipynb`) explores each cleaned dataset's shape, distributions, trends and relationships across 15 charts. Every takeaway was checked against the notebook's executed outputs. Six key charts follow.

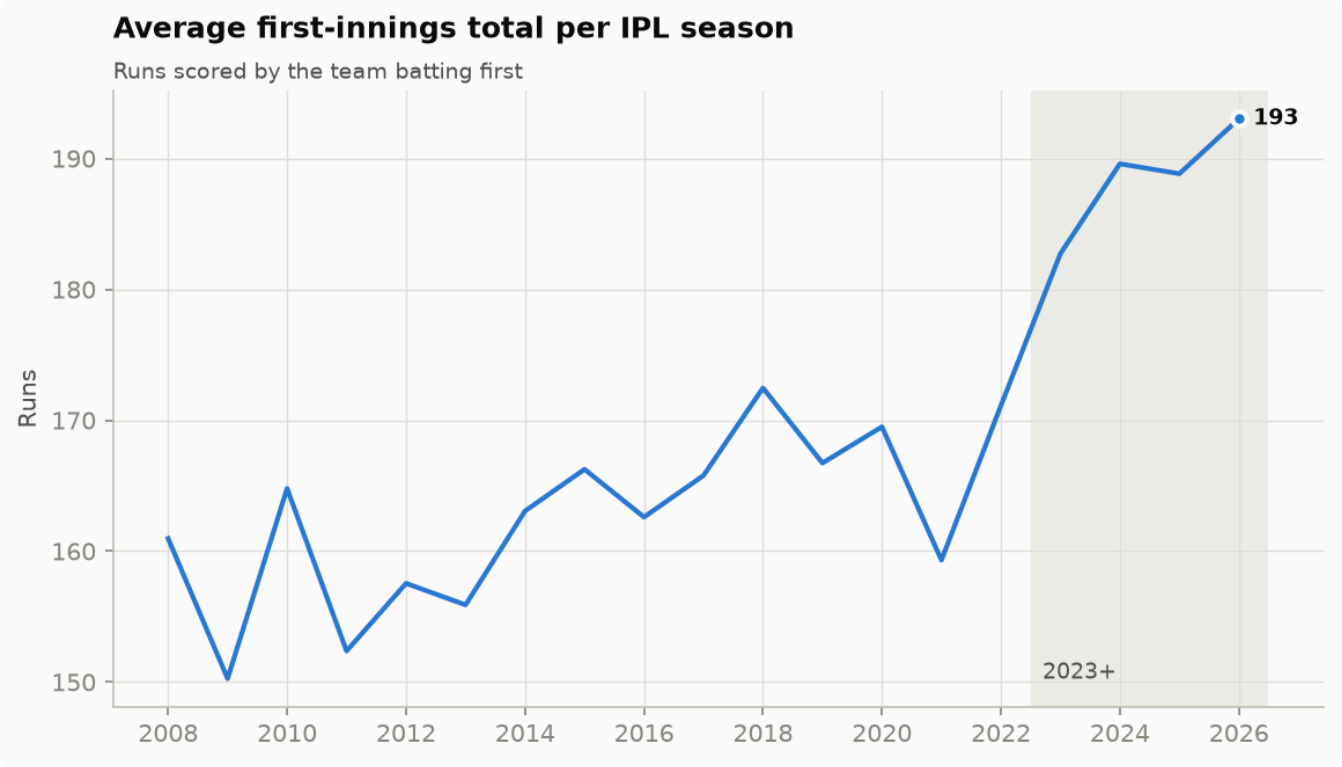
4.1 Olist: late deliveries drive bad reviews

Orders delivered early average about 4.2–4.3 stars. Deliveries 1–3 days late drop to 3.3, and a week or more late to under 2. Late orders account for 37% of all 1-star reviews.

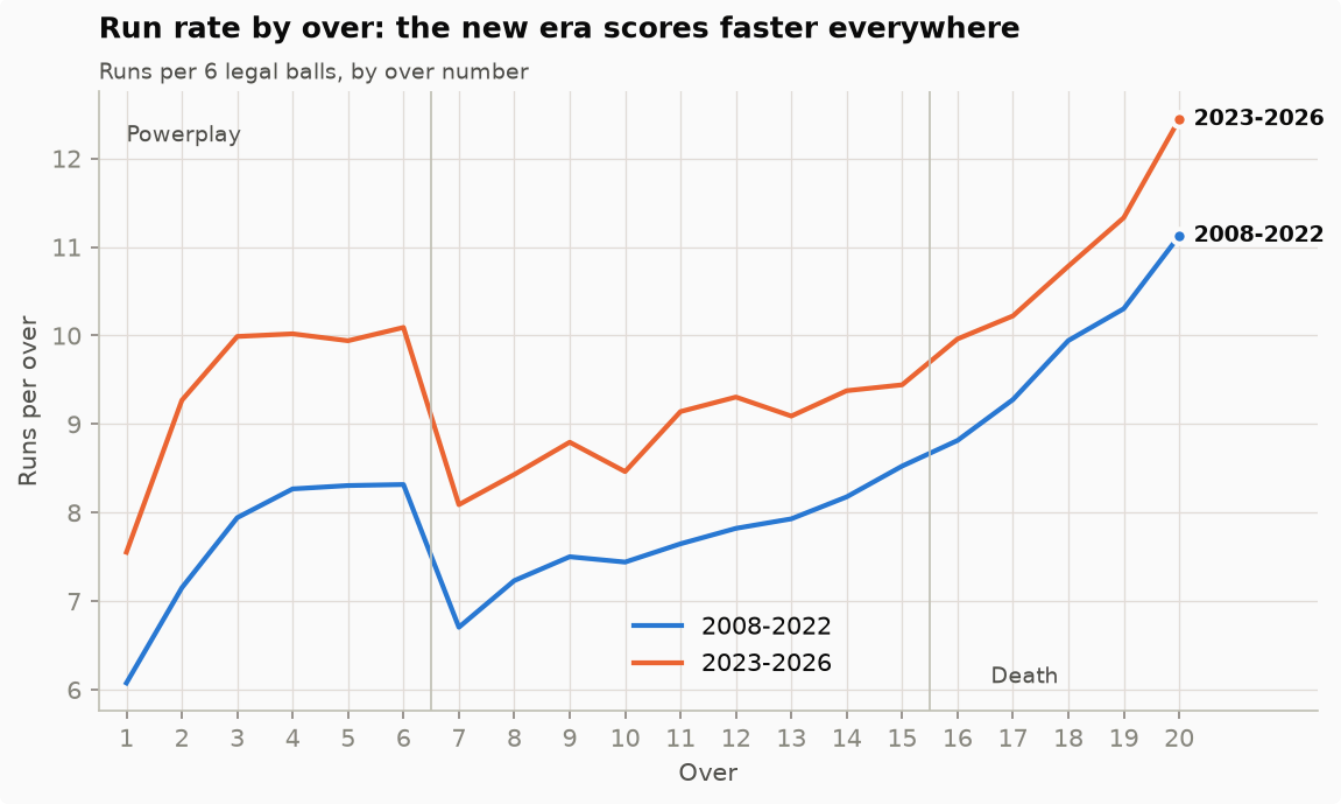


4.2 IPL: scoring has jumped since 2023

Average first-innings totals stayed in a 150–173 band for most of the league's history, then rose to about 190 from 2023.

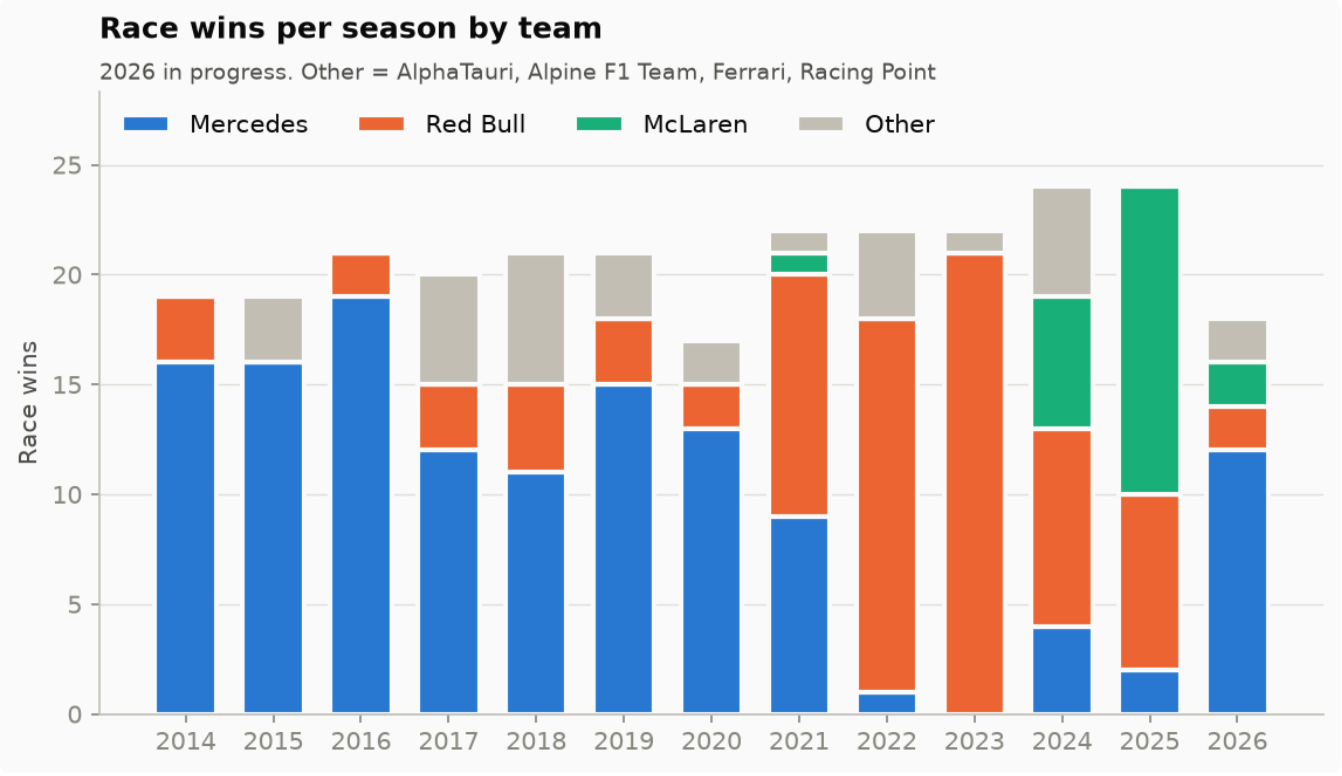


The increase is visible in every phase of the innings, not only at the end: the powerplay run rate rose from 7.7 to 9.5 runs per over.

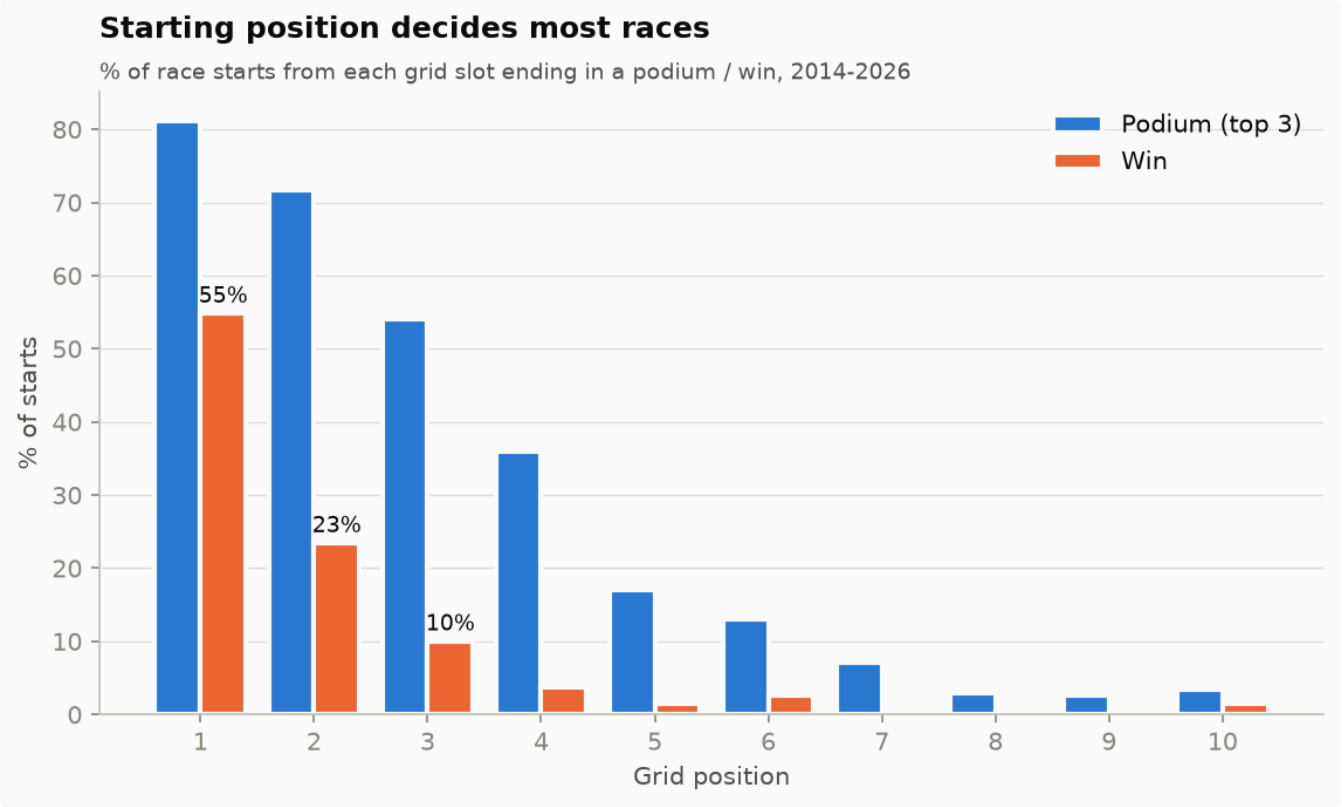


4.3 Formula 1: dominant eras and the value of pole

Mercedes won 102 of 138 races between 2014 and 2020, after which Red Bull and then McLaren led.

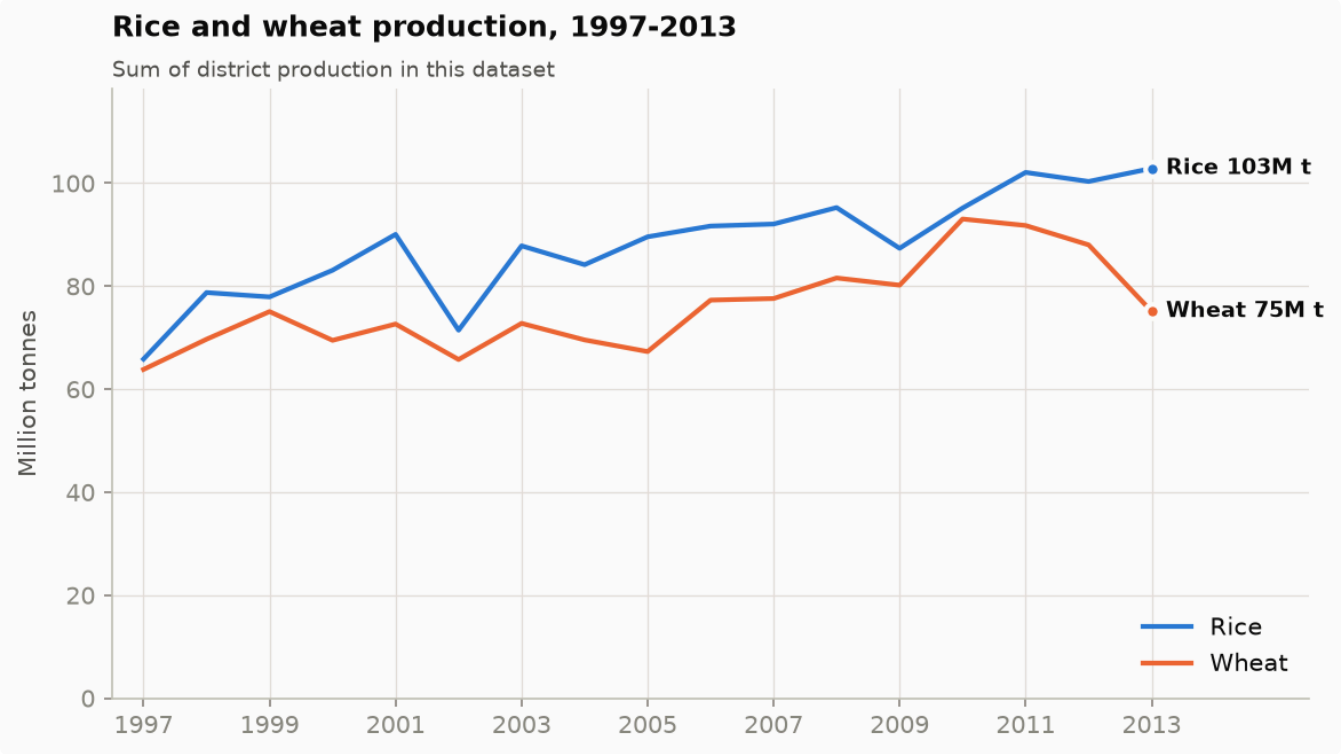


Starting position decides most races: pole position converts to a win 55% of the time, and a podium about 80% of the time.



4.4 Crop production: long-term rice and wheat trends

Rice production rose from 66 million to 103 million tonnes between 1997 and 2013, with a sharp dip in the 2002 drought year. Area explains production well (r = 0.97 on log scales), and yields vary widely between states: Punjab reaches about 3.7 t/ha of rice compared with 1.3–1.6 t/ha in several eastern states.



4.5 Caveats recorded in the notebook

- Olist 2016 and September–October 2018 are partial periods, and the F1 2026 season is still in progress.
- Crop totals across different crops are not meaningful because units differ, and data after 2013 is incomplete.
- F1 pit-stop times include red-flag stoppages (one lasts 3,069 seconds), so medians are used instead of means.

5. Evaluation Question Sets

BIRD tests SQL generation but does not cover CSV files, Hinglish, or questions that should be refused. Four question sets were therefore built on the cleaned data and committed to `data/questions/`.

Set	Rows	Purpose
Answerable (English)	200 (50 per dataset)	SQL accuracy on CSV and SQLite data; difficulty easy 60, medium 77, hard 63
Hinglish (answerable)	100 (25 per dataset)	English vs Hinglish accuracy, linked to the English originals
Unanswerable (English)	200 (10 per type per dataset)	Training and testing the answerability classifier
Hinglish (unanswerable)	50	Ensures Hinglish appears in both classes

Each row stores the question, gold SQL, the gold answer produced by running that SQL, difficulty, and a train or test split.

Unanswerable types. Questions are balanced across five types: *missing column* (the data does not contain it, such as customer age), *out of range* (years or entities absent from the data, verified individually), *external knowledge* (such as a team principal's name), *subjective* (opinions such as "the best season for farming"), and *write request* (such as "Delete all canceled orders").

Quality controls

- A validation script re-runs every gold SQL and confirms the stored answer, row counts, splits and links between Hinglish and English rows. All 200 gold queries ran successfully.

- Gold answers that depended on suspicious data were reviewed. Three questions were replaced (one built on a likely data-entry error, one on a pit-lane timing artifact, one whose answer flipped depending on rain-affected matches) and one was narrowed to 1997–2013.
- All 100 Hinglish questions were reviewed by a native speaker. 15 (15%) were reworded for naturalness, including one whose translation had changed the meaning of the question.

Train/test split and leak protection. Questions are split 70/30, stratified by dataset, answerability and unanswerable type, with a fixed seed. Hinglish rows inherit the split of their English original, so the same gold SQL can never appear in both train and test. A dedicated loader returns only training rows for prompt examples and training, and a unit test enforces that no test question can leak.

6. Feature Engineering

The answerability classifier (trained in Module 4) needs numeric features that describe how well a question fits a dataset. `src/datacopilot/pipeline/features.py` extracts them from the question, the schema, and a dataset context that adds the year range and the vocabulary of values in text columns.

Feature groups

- **Schema fit:** share of question words matching table or column names, and the number of unmatched content words.
- **Value fit:** entity mentions not found in the data, years mentioned, and whether they fall outside the dataset's year range.
- **Intent cues:** write verbs (delete, update, drop), opinion words (best, should), and external-knowledge cues (current, CEO, principal).
- **Question form:** question type (which/what, how many, who, yes/no, imperative), length, numbers mentioned, and the share of Hinglish words.

Signal on the English training split (140 questions per class)

Feature	Answerable	Unanswerable
Schema overlap	0.56	0.29
Unmatched content words	0.82	1.54
Starts with a write verb	0.00	0.20
External-knowledge cues	0.01	0.21

Feature	Answerable	Unanswerable
Opinion words	0.02	0.21
Unmatched entities	0.00	0.18
Year outside data range	0.00	0.09

Schema overlap and unmatched words separate the classes broadly, while the other features each isolate one unanswerable type almost perfectly. 37 of the 40 out-of-range questions are caught by the year or entity features, and no answerable question triggers either.

Hinglish rows (training split). After adding 50 Hinglish unanswerable questions, the feature table covers all 550 questions (385 in training). Schema overlap still separates the classes in Hinglish, while the Hinglish-word share identifies the language rather than answerability:

Training rows	n	Schema overlap: answerable	Schema overlap: unanswerable	Hinglish share: answerable	Hinglish share: unanswerable
English	280	0.56	0.29	0.00	0.00
Hinglish	105	0.50	0.28	0.49	0.41
All	385	0.54	0.29	0.16	0.08

Hinglish intent cues also work: all 6 Hinglish write requests in training ("… kar do", "hata do") are flagged as imperative, and all 8 subjective questions contain an opinion word.

Fixes made after inspecting real rows. A quantity such as "at least 2000 balls" was first read as the year 2000, and weekday names were counted as unknown entities. Both were corrected. Rare outlier dates (4 of 112,000 Olist rows) were excluded when computing year ranges.

Guarding against shortcut learning. Because the same process wrote the unanswerable questions and the features, some features may reflect writing style rather than answerability. `ends_with_question_mark` was removed for this reason, leaving 24 features, and question length is flagged as a possible artifact (unanswerable questions average about 7 words against 10). Hinglish unanswerable questions were added so Hinglish appears in both classes; a remaining 2:1 imbalance (100 answerable vs 50 unanswerable Hinglish questions) is documented and will be checked in Module 4. The label column (`unanswerable_type`) is kept for error analysis only and never used in training. Module 4 will also test the classifier on a separate set of questions written by a person, to measure how well it generalises beyond generated phrasing.

7. Early Pipeline Findings

A working prototype (CSV or SQLite → LLM-generated SQL → validator → execution) was built ahead of schedule and run on the first 10 BIRD dev questions with `qwen2.5-coder:7b` on a local GPU. This is a smoke test, not a benchmark result: all 10 questions use one database.

Mode	Execution accuracy
Baseline (one attempt, no schema check)	6/10 (60%)
Validated (schema check + one retry)	7/10 (70%)

Error log so far

- **Silent wrong column (fixed by the validator).** The model referenced a column from the wrong table. SQLite treats an unknown double-quoted name as a text string, so the query "succeeded" and returned the column name itself as data. The schema check caught it and the retry produced a correct answer.
- **Identical SQL on retry.** At temperature 0 the model regenerated the same invalid SQL even when given the exact error message. A slightly higher temperature on the retry, with a fixed seed, will be tested in Module 4.
- **Valid SQL, wrong meaning.** Two answers used the wrong column or table for the concept asked. No structural check can catch these.
- **Right answer, wrong logic.** Asked which state produced the most rice in 2013, the model ordered individual district rows instead of summing by state. It still returned the correct state (West Bengal), so execution accuracy marks it correct. Correct results can therefore hide incorrect reasoning, and a sample of answers will be inspected manually in Module 4.
- **Instructions not followed.** Asked to return only SQL, the model added explanations, which confirmed the need for a SQL extraction step.

Validator testing also revealed that `SELECT … INTO` could create tables; it is now blocked. Set operations (UNION, INTERSECT, EXCEPT) are allowed only when every branch is a read-only SELECT.

8. Ethics, Privacy and Limitations

Licensing. Olist and F1 are licensed CC BY-NC-SA 4.0 (non-commercial, share-alike), IPL is ODC-By 1.0, and the crop data is published under GODL-India. All are used for academic, non-commercial purposes. Raw files are not redistributed through GitHub or the deployed application, and every source is credited.

Privacy. The datasets contain no directly identifying personal data: Olist customers are anonymised IDs, and the sports and agricultural data are public records. In the deployed app, uploaded user files will be private to each session and deleted afterwards, and the LLM will receive only the schema and a few sample values, not full datasets.

Limitations

- The crop data comes from an unofficial mirror, ends in 2015, and has no stated units.
- Gold answers reflect the cleaned data, including its remaining quirks; if cleaning changes, the answers must be rebuilt and reviewed.
- The answerable and unanswerable questions were generated with AI assistance and then reviewed. They may not fully represent how real users phrase questions, which is why a separate human-written test set will be used.
- Execution accuracy can mark an answer correct even when the SQL logic is wrong.

9. Next Steps (Module 4)

- Train and compare answerability classifiers (Logistic Regression and XGBoost) on the training split, and evaluate on the held-out test split and a human-written question set.
- Run the full BIRD dev set (1,534 questions) and the custom question sets in baseline and validated modes, with statistical testing (McNemar's test).
- Compare local models (Qwen2.5-Coder 7B, Llama 3.1 8B) with a larger model (Qwen2.5-Coder 14B on a cloud notebook GPU) and a hosted model.
- Measure English versus Hinglish accuracy, and test a higher retry temperature.

References

1. Li, J., et al. (2023). *Can LLM Already Serve as a Database Interface? A BIg Bench for Large-Scale Database Grounded Text-to-SQLs.* NeurIPS 2023.
2. Olist. *Brazilian E-Commerce Public Dataset by Olist.* Kaggle. CC BY-NC-SA 4.0.
3. Cricsheet. *IPL ball-by-ball data.* cricsheet.org. Open Data Commons Attribution License 1.0.
4. Jolpica-F1. *Formula 1 database dumps.* github.com/jolpica/jolpica-f1. CC BY-NC-SA 4.0.
5. Ministry of Agriculture and Farmers Welfare. *District-wise, season-wise crop production statistics from 1997.* data.gov.in. Government Open Data License – India. Accessed via the Kaggle mirror "Crop Production in India".
6. McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2), 153–157.